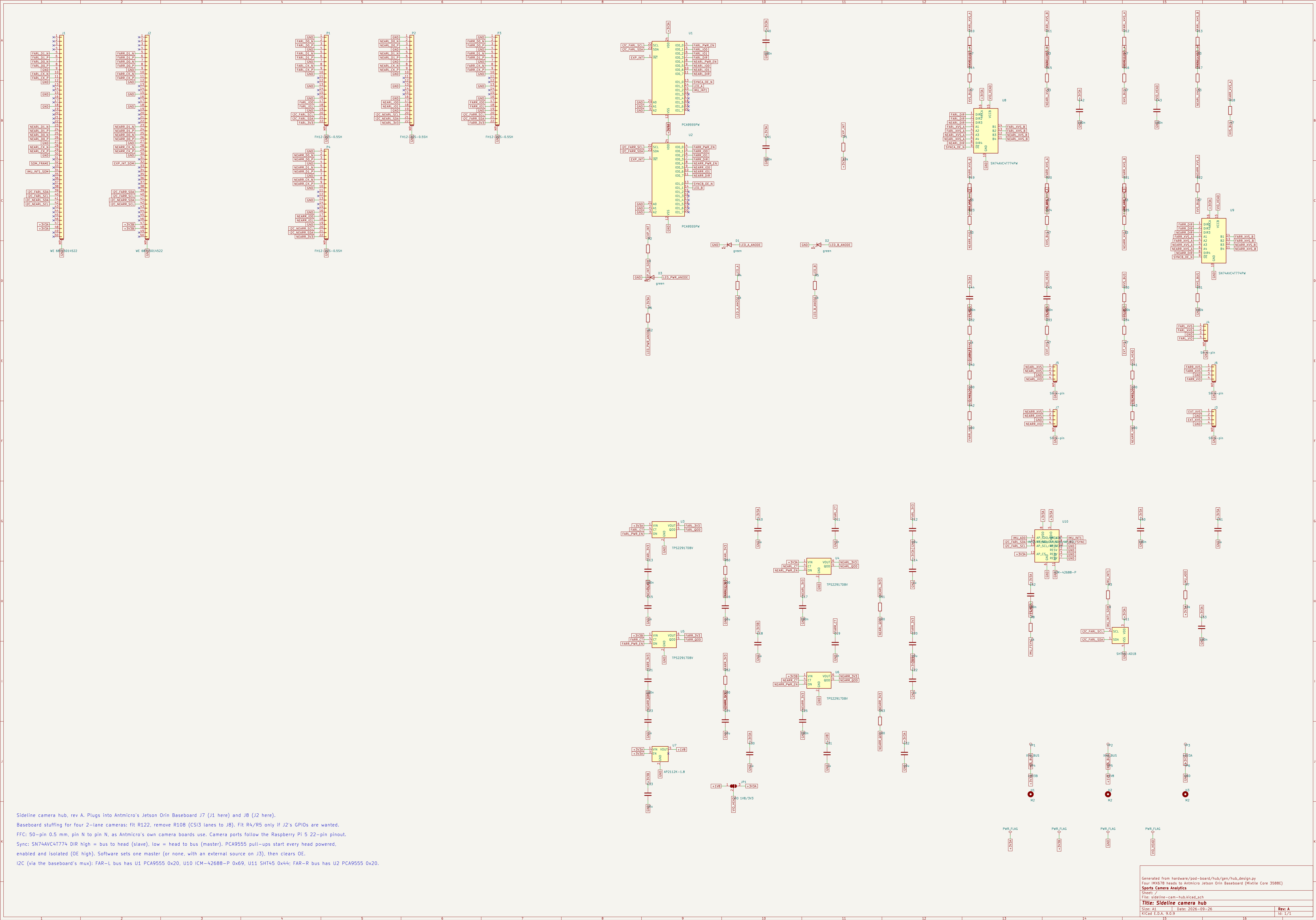




Sideline camera hub, rev A. Plugs into Antmicro's Jetson Orin Baseboard J7 (J1 here) and J8 (J2 here).

Baseboard stuffing for four 2-lane cameras: fit R122, remove R108 (CSI3 lanes to J8). Fit R4/R5 only if J2's GPIOs are wanted.

FFC: 50-pin 0.5 mm, pin N to pin N, as Antmicro's own camera boards use. Camera ports follow the Raspberry Pi 5 22-pin pinout.

Sync: SN74AVC4774 DIR high = bus to head (slave), low = head to bus (master). PCA9555 pull-ups start every head powered,

enabled and isolated (OE high). Software sets one master (or none, with an external source on J3), then clears OE.

I2C (via the baseboard's mux): FAR-L bus has U1 PCA9555 0x20, U10 ICM-42688-P 0x69, U11 SHT45 0x44; FAR-R bus has U2 PCA9555 0x20.